

DOOB'S PROOF OF MLE CONSISTENCY

OMAR HIJAB

ABSTRACT. A concise presentation of Doob's proof of the consistency of Fisher's maximum likelihood estimator.

1. INTRODUCTION.

The consistency of Fisher's maximum likelihood estimator (MLE) [10] is a basic result discovered more than a century ago. This method is typically subsumed under the rubric of M -estimators, and derived via standard techniques dating back to Cramér [6] and Wald [24]. For a more recent treatment, van der Vaart [23] is standard. Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found Doob's consistency proof [9] of MLE convergence more motivated and accessible to students, especially to students already exposed to information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The aim of this note is to present this proof, together with some historical terminology remarks, in a self-contained manner. By using the law of large numbers in its strong form, we derive MLE consistency in its strong form.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2. SURPRISAL AND INFORMATION

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X . We suspect the density of X is $p(X, \theta^*)$, one of a parametric family $p(X, \theta)$ of densities, and we wish to recover θ^* from the samples $X_1, X_2, \dots$.

Stated in this manner, all we have to work with are two items: the true expectation E_* and the family $p(X, \theta)$. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over or knowledge of E_* . Given this, how do we use the samples $X_1, X_2, \dots$ to recover θ^* ?

Define the *surprisal* [18, 21]

$$S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to characterize θ^* as a minimizer of the expected surprisal $E_*(S(X, \theta))$. More precisely, we say θ^* is the *true parameter* if $S(X, \theta)$ is integrable for all θ , and θ^* is the *unique* minimizer of $E_*(S(X, \theta))$. In particular, if θ^* is the true parameter and $\theta \neq \theta^*$, then $p(X, \theta) \neq p(X, \theta^*)$.

Equivalently, given θ^* , the *relative information* is

$$(1) \quad I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)).$$

Then θ^* is the true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ , with $I(\theta^*, \theta) = 0$ only if $\theta = \theta^*$. The relative information I appears in many fields, in different forms. Because of widely differing terminology, below we give an account of its history.

A special case is when there is¹ a background expectation E such that

$$E_\theta(f(X)) = E(p(X, \theta)f(X))$$

¹While $E_\theta(1) = 1$, E need not satisfy $E(1) = 1$, $E(1) = \infty$ is allowed.

is an expectation for all θ . When this holds and $E_{\theta^*} = E_*$, the relative information has the familiar form

$$I(\theta^*, \theta) = E \left(p(X, \theta^*) \log \left(\frac{p(X, \theta^*)}{p(X, \theta)} \right) \right),$$

$I(\theta^*, \theta) \geq 0$ for all θ , and $I(\theta^*, \theta) = 0$ implies² $p(X, \theta) = p(X, \theta^*)$ [7, 8]. When $\theta \neq \theta^*$ implies $p(X, \theta) \neq p(X, \theta^*)$, it follows θ^* is the true parameter.

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of X is $p(X, \theta^*) = \theta_X^*$. If θ is any other probability vector, the surprisal is $S(X, \theta) = \log(1/\theta_X)$, and

$$(2) \quad I(\theta^*, \theta) = \sum_{i=1}^d \theta_i^* \log(\theta_i^*/\theta_i).$$

Turning back to the general setting, the *likelihood* and *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^n p(X_k, \theta), \quad \bar{S}_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^n S(X_k, \theta).$$

The mean surprisal is also called the negative-log-likelihood.

By definition, θ^* minimizes $E_*(S(X, \theta))$. By the law of large numbers, $\bar{S}_n(\theta)$ converges to $E_*(S(X, \theta))$ as $n \rightarrow \infty$, with probability one. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ . Under **A** and **B** below, with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n .

3. CONSISTENCY

We make two assumptions. A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c . Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the first assumption

A. (Properness): There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,

$$(3) \quad F(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N.$$

We will need the convergence of $\bar{S}_n(\theta)$ to $E_*(S(X, \theta))$ across θ , in the sense

$$(4) \quad \bar{S}_n(\theta) \rightarrow E_*(S(X, \theta)), \quad \text{as } n \rightarrow \infty, \quad \text{for all } \theta,$$

with probability one. For this we need the second assumption

B. (Continuity): The parameter space is separable and there is a continuous $f(\theta, \theta')$ and an integrable $g(X)$ satisfying

$$|S(X, \theta) - S(X, \theta')| \leq f(\theta, \theta')g(X) \quad \text{and} \quad f(\theta, \theta) = 0.$$

Then a separability argument [19] establishes (4) with probability one.

Under **A** and **B**, an MLE $\hat{\theta}_n$ exists³ for $n \geq N$, with probability one. **A** and **B** are typically checkable for *exponential families* [5, 15, 19, 23], the densities $p(X, \theta)$ where $S(X, \theta)$ is a sum of products of the form $f(\theta)g(X)$.

Theorem (Consistency). *Let $X_1, X_2, \dots$ be independent and identically distributed, and let $p(X, \theta)$ be a parametric family with true parameter θ^* . Under **A** and **B**, with probability one, an MLE sequence $\hat{\theta}_n$ may be defined for sufficiently large n , and $\hat{\theta}_n$ converges to θ^* as $n \rightarrow \infty$.*

²Apply Jensen's inequality with the strictly convex $\log(1/p)$.

³A proper lower semicontinuous function has a minimizer.

Proof. By the law of large numbers,

$$(5) \quad \frac{1}{n} \sum_{k=1}^n g(X_k) \rightarrow E_*(g(X)), \quad \text{as } n \rightarrow \infty,$$

with probability one. Fix samples for which (3), (4) and (5) hold. We show $\hat{\theta}_n \rightarrow \theta^*$ for these samples. By (3),

$$F(\hat{\theta}_n) \leq \bar{S}_n(\hat{\theta}_n) \leq \bar{S}_n(\theta^*), \quad \text{for } n \geq N.$$

Since the right side is bounded as $n \rightarrow \infty$, and $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n \rightarrow \theta$ along a subsequence. By B,

$$(6) \quad \bar{S}_n(\theta) - \bar{S}_n(\theta^*) \leq \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \leq f(\theta, \hat{\theta}_n) \cdot \frac{1}{n} \sum_{k=1}^n g(X_k).$$

Since the left side of (5) converges to $E_*(g(X))$, and $f(\theta, \hat{\theta}_n) \rightarrow 0$ along a subsequence, the right side of (6) goes to zero along a subsequence. By (1) and (4), the left side of (6) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \leq 0$. But $I(\theta^*, \theta) \geq 0$, hence $I(\theta^*, \theta) = 0$, showing $\theta = \theta^*$. Thus $\hat{\theta}_n \rightarrow \theta^*$ as $n \rightarrow \infty$. $\square$

4. TERMINOLOGY HISTORY

The relative information $I(\theta^*, \theta)$ is often called “relative entropy” [5, 15]. The term “entropy”, originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann’s 1877 combinatorial interpretation [3] and Planck’s 1901 paper [17], flipped Boltzmann’s sign convention, writing the entropy H as done today.

Shannon, in his 1948 paper [20], independently defined

$$H(p) = - \sum_{i=1}^d p_i \log p_i.$$

Von Neumann had earlier defined quantum entropy in his 1932 book [16], and when Shannon consulted him around 1939–1940 about terminology for H , von Neumann suggested “entropy”, which Shannon adopted [22]. However, in his remark “ H is then, for example, the H in Boltzmann’s H theorem”, Shannon conflates his H with Boltzmann’s H .

The relative information $I(\theta^*, \theta)$ is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, a fact that is apparent in the special case (2), whereas $H(p)$ above is strictly concave in p . Because of this, calling both quantities “entropy” creates a mismatch. Indeed, based on the curvature, it is natural to call the *negative* of (2) the *relative entropy*. The mismatch is also in Python: `entropy(p)` returns the concave H , but `entropy(p, q)` returns the convex I .

This confusion can be dissolved by remembering entropy and information are opposites: entropy is concave and information is convex. More generally, this error is a common psychological conflation: When measuring the temperature, are we measuring how cold it is, or how hot it is? Of course we are doing both, but this doesn’t mean cold equals hot. In fact, cold equals minus hot.

While many authors’ terminology conflicts with ours, some authors align partially with our usage. Schervish, in his 1995 text [19], uses “Kullback-Leibler information” for I , which is more descriptive than the widely used but opaque “Kullback-Leibler divergence” [1]. Kullback himself, in his 1959 monograph [14], refers to I as an “information function”. Donsker-Varadhan, in their 1975 large deviations paper [8], denote their rate function by I rather than H ; the relative information is a special case of their I .

Regarding Doob’s 1934 proof, this was at the dawn of abstract probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of Kolmogorov’s 1933 foundational extension theorem [13].

5. MULTIVARIATE NORMAL

Let μ be in $\mathbf{R}^d$, Q a positive definite symmetric $d \times d$ matrix, and $\theta = (\mu, Q)$. Let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(X, \theta) = e^{-S(X, \theta)}$ where

$$(7) \quad S(X, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} (X - \mu) \cdot Q^{-1} (X - \mu).$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter $\theta^* = (\mu^*, Q^*)$* if its moment-generating function is

$$(8) \quad E_*(e^{v \cdot X}) = \exp \left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v \right).$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (8), and let $X_1, X_2, \dots$ be independent and identically distributed copies⁴ of X . We show θ^* is the true parameter, and we verify **A** and **B**.

Assumption **B** follows from (7). To establish **A**, we review some matrix manipulations. The proofs of the Lemmas below are in the Appendix.

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v \otimes w$ is the matrix whose ij -th entry is $v_i w_j$. Then $v \otimes v$ is symmetric,

$$(9) \quad v \cdot Qv = \text{trace}(Q(v \otimes v)), \quad v \otimes w + w \otimes v \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w,$$

and, by the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v \otimes v$.

Given symmetric matrices Q, R, S , we say $Q \geq 0$ if $v \cdot Qv \geq 0$ for all v , and $Q \geq R$ if $Q - R \geq 0$. If $Q \geq \delta I$ for some $\delta > 0$, we say $Q > 0$. If $Q \geq 0$ and $R \geq S$, then $\text{trace}(QR) \geq \text{trace}(QS)$ and $\text{trace}(QR) = \text{trace}(RQ)$.

The determinant $\det(Q)$ is the product of the eigenvalues of Q .

Lemma 1. *With Q invertible symmetric and t a scalar,*

$$(10) \quad (Q + tv \otimes v)^{-1} = Q^{-1} - \frac{t}{1 + tv \cdot Q^{-1}v} \cdot (Q^{-1}v) \otimes (Q^{-1}v),$$

as long as $1 + tv \cdot Q^{-1}v \neq 0$.

Lemma 2. *With Q invertible symmetric and t a nonzero scalar,*

$$(11) \quad \det(Q + tv \otimes v) = \det(Q) (1 + tv \cdot Q^{-1}v).$$

Lemma 3. *If $S > 0$ and $Z > 0$, then $\det(SZS) = \det(Z) \det(S)^2$.*

Lemma 4. *Let $h(\lambda) = \log \lambda + 1/\lambda$ for $\lambda > 0$. Then $h(\lambda) \geq 1$ and $h(\lambda) = 1$ only if $\lambda = 1$, and $h(\lambda) \leq c$ implies $e^{-c} \leq \lambda \leq e^c$.*

Lemma 5. *Let $f(\mu)$ be a symmetric matrix-valued function such that $\text{trace}(f(\mu))$ is proper and $f(\mu) \geq \delta I$ for some $\delta > 0$. Then*

$$(12) \quad F(\mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1} f(\mu)).$$

is proper.

Let $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$. In (8), multiply both sides by $e^{-v \cdot \mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$(13) \quad E_*(X) = \mu^*, \quad E_*((X - \mu) \otimes (X - \mu)) = f(\mu).$$

By (13), $S(X, \theta)$ is integrable, $E_*(S(X, \theta)) = F(\mu, Q)$, and

$$I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)) = F(\mu, Q) - F(\mu^*, Q^*).$$

Since $\text{trace}(f(\mu)) = \text{trace}(Q^*) + |\mu^* - \mu|^2$, $\text{trace}(f(\mu))$ is proper. Moreover, $f(\mu) \geq \delta I$ for some $\delta > 0$. By Lemma 5, $F(\mu, Q)$ is proper, hence $E_*(S(X, \theta))$ is proper, hence $I(\theta^*, \theta)$ is proper.

⁴This setup is a special case of the Kolmogorov extension theorem.

Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1}QS^{-1}$ and let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Z . Then $\text{trace}(Q^{-1}Q^*) = \text{trace}(SQ^{-1}S) = \text{trace}(Z^{-1})$. Since $f(\mu) \geq Q^*$, by Lemmas 3 and 4,

$$\begin{aligned} 2I(\theta^*, \theta) &\geq \log \det(SZS) - \log \det(S)^2 + \text{trace}(Z^{-1} - I) \\ &= \log \det Z + \text{trace}(Z^{-1} - I) = \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0. \end{aligned}$$

If this chain of inequalities is an equality, then $Z = I$ and $\mu = \mu^*$. Hence θ^* is the true parameter.

Let $0 < \epsilon < 1/2$ and let $f_\epsilon(\mu) = (1 - \epsilon)f(\mu) - \epsilon Q^*$. By Lemma 5 again,

$$F_\epsilon(\theta) = F_\epsilon(\mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1}f_\epsilon(\mu))$$

is proper.

Let $\bar{X}_n = (X_1 + \dots + X_n)/n$ be the sample mean. By the law of large numbers and (13) with $\mu = \mu^*$, there is $N \geq 1$, depending on the samples, such that, with probability one,

$$(\bar{X}_n - \mu^*) \otimes (\bar{X}_n - \mu^*) \leq \epsilon^2 Q^*, \quad \text{for } n \geq N,$$

and

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu^*) \otimes (X_k - \mu^*) \geq (1 - \epsilon)Q^*, \quad \text{for } n \geq N.$$

With $w = \bar{X}_n - \mu^*$ and $v = \mu - \mu^*$, using (9),

$$v \otimes w + w \otimes v \leq \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\bar{X}_n - \mu^*) \otimes (\bar{X}_n - \mu^*) \leq \epsilon f(\mu)$$

for all μ and $n \geq N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu) - \frac{1}{n} \sum_{k=1}^n (X_k - \mu^*) \otimes (X_k - \mu^*)$$

equals $v \otimes v - (v \otimes w + w \otimes v)$. Combined with the last two inequalities, this leads to

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu) \geq f_\epsilon(\mu)$$

for all μ and $n \geq N$. Since this implies

$$F_\epsilon(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N,$$

this establishes A.

By differentiating $\bar{S}_n(\mu + tv, Q)$ and $\bar{S}_n(\mu, Q + tv \otimes v)$ at $t = 0$, the MLE is

$$\hat{\mu}_n = \frac{1}{n} \sum_{k=1}^n X_k, \quad \hat{Q}_n = \frac{1}{n} \sum_{k=1}^n (X_k - \hat{\mu}_n) \otimes (X_k - \hat{\mu}_n).$$

6. DISCLOSURES

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A. PROOFS OF LEMMAS

Proof of Lemma 1. Direct calculation. □

Proof of Lemma 2. If v is orthogonal to an eigenvector x of Q , then we reduce to a lower-dimensional case by restricting to $x^\perp$. Hence we may assume $v \cdot x \neq 0$ for all eigenvectors x of Q . If v is orthogonal to an eigenvector x' of $Q' = Q + tv \otimes v$, then x' is an eigenvector of Q , contradicting our assumption. Hence we may assume $v \cdot x' \neq 0$ for all eigenvectors x' of Q' . If x'_1 and x'_2 are linearly independent eigenvectors of Q' corresponding to the same eigenvalue λ , then, for suitable r_1, r_2 , $x' = r_1 x'_1 + r_2 x'_2$ is an eigenvector of Q'

satisfying $v \cdot x' = 0$, contradicting our assumptions. Hence we may assume the eigenvalues of Q' are simple. If x and x' are eigenvectors for Q and Q' corresponding to the same eigenvalue λ , then

$$t(v \cdot x')(v \cdot x) = x' \cdot (tv \otimes v)x = x' \cdot Q'x - x' \cdot Qx = 0,$$

contradicting our assumptions. Hence we may assume the eigenvalues of Q and the eigenvalues of Q' do not overlap. If λ is an eigenvalue of Q' with eigenvector x' , then

$$\begin{aligned} (v \cdot x') (1 + t(v \cdot (Q - \lambda I)^{-1}v)) &= v \cdot (x' + t(v \cdot x')(Q - \lambda I)^{-1}v) \\ &= (Q - \lambda I)^{-1}v \cdot ((Q - \lambda I)x' + t(v \cdot x')v) \\ &= (Q - \lambda I)^{-1}v \cdot (Q' - \lambda I)x' = 0, \end{aligned}$$

hence λ is a root of the polynomial $p(\lambda) = \det(Q - \lambda I)(1 + tv \cdot (Q - \lambda I)^{-1}v)$. Thus the eigenvalues of Q' are the roots of $p(\lambda)$. Since the product of the roots of $p(\lambda)$ is $p(0)$, the result follows. $\square$

Proof of Lemma 3. Assume the result holds for Z . By (11),

$$\begin{aligned} \det(S(Z + v \otimes v)S) &= \det(SZS + Sv \otimes Sv) \\ &= \det(SZS)(1 + Sv \cdot (SZS)^{-1}Sv) \\ &= \det(Z) \det(S)^2 (1 + v \cdot Z^{-1}v) = \det(Z + v \otimes v) \det(S)^2, \end{aligned}$$

hence the result holds for $Z + v \otimes v$. If $Z = \delta I$, the result holds. Since every $Z > 0$ is a sum of δI and a sum of matrices of the form $v \otimes v$, the result follows by induction over the number of terms in the sum. $\square$

Proof of Lemma 4. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \leq c$ implies $\lambda \leq e^c$. Since $\lambda \log \lambda \geq -1/e$ and $(e - 1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1 - 1/e)/c \geq e^{-c}$. $\square$

Proof of Lemma 5. Since $F(\mu, Q)$ is proper iff $F(\mu, \delta Q)$ is proper, and, up to a constant, $F(\mu, \delta Q)$ corresponds to $f(\mu)/\delta$, we may assume $\delta = 1$. Now suppose $2F(\mu, Q) \leq c$. It is enough to establish $\epsilon I \leq Q \leq I/\epsilon$ for $\epsilon = e^{-c}$, since this confines Q to a compact subset of the space of positive definite matrices, and implies

$$c \geq 2F(\mu, Q) \geq \log \det(2\pi\epsilon I) + \epsilon \operatorname{trace}(f(\mu)),$$

which confines μ to a compact subset of $\mathbf{R}^d$.

Let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Q . Since $f(\mu) \geq I$,

$$(14) \quad \sum_{i=1}^d h(\lambda_i) = \log \det(Q) + \operatorname{trace}(Q^{-1}) \leq 2F(\mu, Q) - \log \det(2\pi I) \leq c.$$

Since $h(\lambda) \geq 1$, for each i , $h(\lambda_i) \leq c$, hence $e^{-c} \leq \lambda_i \leq e^c$, establishing $\epsilon I \leq Q \leq I/\epsilon$. $\square$

REFERENCES

- [1] Bishop, C.M.: Pattern Recognition and Machine Learning. Information Science and Statistics. Springer, Cham (2006)
- [2] Boltzmann, L.: Weitere Studien über das Wärmegleichgewicht unter Gasmolekülen. Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften. Mathematisch-Naturwissenschaftliche Classe. Abteilung II **66**, 275–370 (1872)
- [3] Boltzmann, L.: Über die beziehung zwischen dem zweiten hauptsatze der mechanischen wärmetheorie und der wahrscheinhlichkeitsrechnung respektive den sätzen über das wärmegleichgewicht. Wiener Berichte **76**, 373–435 (1877)
- [4] Clausius, R.: Über verschiedene für die anwendung bequeme formen der hauptgleichungen der mechanischen wärmetheorie. Annalen der Physik und Chemie **125**, 353–400 (1865)
- [5] Cover, T.M., Thomas, J.A.: Elements of Information Theory, 2nd edn. Wiley-Interscience (2006)
- [6] Cramér, H.: Mathematical Methods of Statistics. Princeton University Press (1946)
- [7] Dembo, A., Zeitouni, O.: Large Deviations Techniques and Applications, *Stochastic Modelling and Applied Probability*, vol. 38, 2nd edn. Springer, New York (1998)
- [8] Donsker, M.D., Varadhan, S.R.S.: Asymptotic evaluation of certain markov process expectations for large time, I. Communications on Pure and Applied Mathematics **28**(1), 1–47 (1975)
- [9] Doob, J.L.: Probability and statistics. Transactions of the American Mathematical Society **36**, 759–775 (1934)
- [10] Fisher, R.A.: On the mathematical foundations of theoretical statistics. Philosophical Transactions of the Royal Society of London, Series A **222**, 309–368 (1922)
- [11] Gibbs, J.W.: Elementary Principles in Statistical Mechanics. Yale University Press, New Haven (1902)
- [12] Hijab, O.: Math for Data Science. Springer, Cham (2025)
- [13] Kolmogorov, A.N.: Grundbegriffe der Wahrscheinlichkeitsrechnung. Springer, Berlin (1933)
- [14] Kullback, S.: Information Theory and Statistics. Wiley, New York (1959)

- [15] Murphy, K.P.: Probabilistic Machine Learning: An Introduction. MIT Press (2022)
- [16] von Neumann, J.: Mathematische Grundlagen der Quantenmechanik. Springer, Berlin (1932)
- [17] Planck, M.: Ueber das gesetz der energieverteilung im normalspectrum. *Annalen der Physik* **309**(3), 553–563 (1901)
- [18] Samson, E.W.: Fundamental natural concepts of information theory. Report E5079, Communications Laboratory, Electronics Research Division, Air Force Cambridge Research Center, Cambridge, MA (1951)
- [19] Schervish, M.J.: Theory of Statistics. Springer, New York (1995)
- [20] Shannon, C.E.: A mathematical theory of communication. *Bell System Technical Journal* **27**(3), 379–423 (1948)
- [21] Tribus, M.: Thermostatistics and Thermodynamics: An Introduction to Energy, Information and States of Matter, With Engineering Applications. D. Van Nostrand Company, Princeton, NJ (1961)
- [22] Tribus, M., McIrvine, E.C.: Energy and information. *Scientific American* **225**(3), 179–190 (1971)
- [23] van der Vaart, A.W.: Asymptotic Statistics, *Cambridge Series in Statistical and Probabilistic Mathematics*, vol. 3. Cambridge University Press, Cambridge (1998)
- [24] Wald, A.: Note on the consistency of the maximum likelihood estimate. *Annals of Mathematical Statistics* **20**(4), 595–601 (1949)